

NIM : 2802490946

Identity & Basic Info

Customer Name

Beatrice

Age

32

Month

January

Occupation

Doctor

Income & Debt Banking

Annual Income

1000000000

Monthly inhand Salary

1000000.00

Number of Bank Accounts

29

Number of Credit Cards

2

Interest Rate (%)

18

Number of Loans

0

Type of Loans

Credit-Builder Loan, and Mortgage Loan

Outstanding Debt

0

Credit Behavior History

Delay from Due Date (Days)

0

Number of Delayed Payments

0

Changed Credit Limit (Raw String)

0

Number of Credit Inquiries

4.00

Credit Mix

Good

Credit Utilization Ratio

32.01

Credit History Age

10 Years and 10 Months

Payment of Minimum Amount

NM

Total EMI per Month

81.82

Amount Invested Monthly (Raw String)

100000

Payment Behaviour

Low_spent_Large_value_payments

Monthly Balance (Raw String)

0

Analyze Credit Score

Prediction Result

Result: Good Kredit (Nasabah Sangat Aman)

Class Probabilities Graph:

 Identity & Basic Info

Customer Name
Beatrice

Age
32

Month
January

Occupation
Doctor

 Income & Debt Banking

Annual Income
56125.5

Monthly Inhand Salary
4875.13

Number of Bank Accounts
8

Number of Credit Cards
3

Interest Rate (%)
18

Number of Loans
2

Type of Loans
Credit-Buider Loan, and Mortgage Loan

Outstanding Debt
370.22

 Credit Behavior History

Delay from Due Date (Days)
30

Number of Delayed Payments
14

Changed Credit Limit (Raw String)
17.89

Number of Credit Inquiries
4.00

Credit Mix
Good

Credit Utilization Ratio
32.01

Credit History Age
28 Years and 10 Months

Payment of Minimum Amount
Yes

Total EMI per Month
81.82

Amount Invested Monthly (Raw String)
182.0655

Payment Behaviour
High_spent_Medium_value_payments

Monthly Balance (Raw String)
473.6241

Analyze Credit Score

 Prediction Result

Result: Standard Kredit 🚩 (Nasabah Cukup Aman)

 Identity & Basic Info

Customer Name
Fishes

Age
40

Month
January

Occupation
Doctor

 Income & Debt Banking

Annual Income
1000

Monthly Inhand Salary
50.00

Number of Bank Accounts
8

Number of Credit Cards
10

Interest Rate (%)
29

Number of Loans
100

Type of Loans
Credit-Buider Loan, and Mortgage Loan

Outstanding Debt
100000000

 Credit Behavior History

Delay from Due Date (Days)
1000

Number of Delayed Payments
200

Changed Credit Limit (Raw String)
200

Number of Credit Inquiries
2.00

Credit Mix
Bad

Credit Utilization Ratio
40.00

Credit History Age
40 Years and 10 Months

Payment of Minimum Amount
No

Total EMI per Month

0.00



Amount Invested Monthly (Raw String)

0

Payment Behaviour

Low_spent_Large_value_payments



Monthly Balance (Raw String)

200

Analyze Credit Score

Prediction Result

Result: Poor Kredit ✖ (Nasabah Berisiko Tinggi)

```
# Baris 1: Profil Risiko Tinggi (Sangat "Poor")
[0, "0x45c7", "CUS_0x3969", "February", "Petunia Olaps", 30, "316-61-XXX", "Scientist",
"16154.25", 1000.0, 8, 8, 29, 5, "Credit-Builder Loan, Auto Loan, and Credit-Builder Loan",
31, 20, 4.8, 10.0, "Bad", 1426.76, 35.0, "15 Years and 11 Months", "Yes", 30.0,
116.134, "Low_spent_Large_value_payments", 247.416],
```

--- Ringkasan Hasil Test Case ---
Data Kasus ke-1 menghasilkan Prediksi Kelas: Poor

```
# Baris 2: Profil Standar (Kemungkinan "Standard")
[1, "ID_002", "CUST_002", "January", "Jane Doe", "34", "123-46-XXX", "Teacher",
"50000", 3500.0, 5, 6, 12.0, 3, "Personal Loan", 10.0,
"4", "10.2", 4.0, "Standard", 1500.0, 50.0, "5 Years", "NM", 200.0, "150",
"Medium_spent_Medium_value_payments", "300.0"]],
```

Data Kasus ke-2 menghasilkan Prediksi Kelas: Standard

```
# Baris 3: Profil Bersih/Ideal (Kemungkinan "Good")
[2, "ID_003", "CUST_003", "January", "Alice Smith", "29", "123-47-XXX", "Engineer",
"120000", 8500.0, 2, 3, 5.0, 1, "Home Equity Loan", 1.0,
"0", "2.1", 1.0, "Good", 200.0, 15.2, "12 Years", "No", 100.0, "500",
"High_spent_Medium_value_payments", "1500.0"]
```

Data Kasus ke-3 menghasilkan Prediksi Kelas: Good

Bukti deployment cloud :

The screenshot shows a Jupyter Notebook interface with the following content in the `main_pipeline.py` tab:

```
[ec2-user@ip-172-23-132-215 ~]$ cd SageMaker
[ec2-user@ip-172-23-132-215 SageMaker]$ pip install scikit-learn==1.2.1
ERROR: Ignored the following versions that require a different python version: 1.8.0 Requires-Python >=3.11; 1.8.0rc1 Requires-Python >=3.11; 1.9.0 Requires-Python >=3.11; 1.9.0rc1 Requires-Python >=3.11
ERROR: Could not find a version that satisfies the requirement scikit-learn==1.2.1 (from versions: 0.9, 0.10, 0.11, 0.12, 0.12.1, 0.13, 0.13.1, 0.14, 0.14.1, 0.15.0, 0.15.1, 0.15.2, 0.16.0, 0.16.1, 0.17, 0.17.1, 0.18, 0.18.1, 0.18.2, 0.19.0, 0.19.1, 0.19.2, 0.20.0, 0.20.1, 0.20.2, 0.20.3, 0.20.4, 0.21.1, 0.21.2, 0.21.3, 0.22, 0.22.1, 0.22.2.post1, 0.23.0, 0.23.1, 0.23.2, 0.24.0, 0.24.1, 0.24.2, 1.0, 1.0.1, 1.0.2, 1.1.0, 1.1.1, 1.1.2, 1.1.3, 1.2.0rc1, 1.2.0, 1.2.1, 1.2.2, 1.3.0rc1, 1.3.0, 1.3.1, 1.3.2, 1.4.0rc1, 1.4.0, 1.4.1.post1, 1.4.2, 1.5.0rc1, 1.5.0, 1.5.1, 1.5.2, 1.6.0rc1, 1.6.0, 1.6.1, 1.7.0rc1, 1.7.0, 1.7.1, 1.7.2)
ERROR: No matching distribution found for scikit-learn==1.2.1
[ec2-user@ip-172-23-132-215 SageMaker]$ pip install scikit-learn==1.2.1
Requirement already satisfied: scikit-learn==1.2.1 in /home/ec2-user/anaconda3/envs/JupyterSystemEnv/lib/python3.10/site-packages (1.2.1)
Requirement already satisfied: numpy>=1.17.3 in /home/ec2-user/anaconda3/envs/JupyterSystemEnv/lib/python3.10/site-packages (from scikit-learn==1.2.1) (1.26.4)
Requirement already satisfied: scipy>=1.3.2 in /home/ec2-user/anaconda3/envs/JupyterSystemEnv/lib/python3.10/site-packages (from scikit-learn==1.2.1) (1.15.3)
Requirement already satisfied: joblib>=1.1.1 in /home/ec2-user/anaconda3/envs/JupyterSystemEnv/lib/python3.10/site-packages (from scikit-learn==1.2.1) (1.5.3)
Requirement already satisfied: threadpoolctl>=2.0.0 in /home/ec2-user/anaconda3/envs/JupyterSystemEnv/lib/python3.10/site-packages (from scikit-learn==1.2.1) (3.6.0)
[ec2-user@ip-172-23-132-215 SageMaker]$ python main_pipeline.py
=====
Approach B - sklearn Pipeline
=====

Step 1: Data Ingestion
Data ingested: /home/ec2-user/SageMaker/data/raw/data_D.csv -> /home/ec2-user/SageMaker/data/ingested/data_D.csv

Step 2: Load and Feature Engineering
Feature Engineering is done. :D

Step 3: Split the data
Splitting done
Nilai negatif pada kolom target berhasil
Memproses 29 kolom numerik...
Proses Winsorizing selesai
```

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https://uas-model-deploy.notebook.us-east-1.sagemaker.aws/lab/tree/src

File Edit View Run Kernel Git Tabs Settings Help

pre_processing.py main_pipeline.py ec2-user@ip-172-23-132-21 x deployEndpoint.py

Step 3: Split the data
Splitting done
Nilai negatif pada kolom target berhasil
Memproses 29 kolom numerik...
Proses Winsorizing selesai

Nilai negatif pada kolom target berhasil
Memproses 29 kolom numerik...
Proses Winsorizing selesai untuk semua kolom numerik (Inference Mode)!

Data Setelah Cleaning & Encoding:

Unnamed: 0	ID	Customer_ID	Month	Savings_Ratio	Remaining_Money	Delayed_Payment_Ratio	Inquiry_per_Loan
0	0x20c27	CUS_0x764	June	0.037346	4793.302143	6.999997	1.999999
1	0xc518	CUS_0x697f	March	0.063425	2821.504472	4.999999	1.600000
2	0x22663	CUS_0x7846	February	0.024200	7341.289635	6.333331	1.666666
3	0x1ff7	CUS_0x3ef5	June	0.085197	449.042077	2.833333	1.166666
4	0x9a07	CUS_0x8df	June	0.023856	2853.111544	6.666664	0.666666

[5 rows x 35 columns]

Step 4: Build and Train Pipeline
/home/ec2-user/anaconda3/envs/2uppyterSystemEnv/lib/python3.10/site-packages/sklearn/linear_model/_logistic.py:458: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
<https://scikit-learn.org/stable/modules/preprocessing.html>
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = check_optimize_result(
2026/07/02 23:44:04 WARNING mlflow.models.model: 'artifact_path' is deprecated. Please use 'name' instead.
Registered model 'uas_model1' already exists. Creating a new version of this model...
Created version '4' of model 'uas_model1'.
Logistic Regression trained. Run ID: 6f7385d310fa4059b264ef5c46be6765
2026/07/02 23:44:18 WARNING mlflow.models.model: 'artifact_path' is deprecated. Please use 'name' instead.

Simple 2 0

ec2-user@ip-172-23-132-215-~/SageMaker 1

08:40
03/07/2026

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https://uas-model-deploy.notebook.us-east-1.sagemaker.aws/lab/tree/src

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pre_processing.py main_pipeline.py ec2-user@ip-172-23-132-21 x deployEndpoint.py

Logistic Regression trained. Run ID: 6f7385d310fa4059b264ef5c46be6765
2026/07/02 23:44:18 WARNING mlflow.models.model: 'artifact_path' is deprecated. Please use 'name' instead.
Registered model 'uas_model2' already exists. Creating a new version of this model...
Created version '4' of model 'uas_model2'.
Random Forest trained. Run ID: 5220f33bcaa7436eae144f394c8c5f2
2026/07/02 23:44:26 WARNING mlflow.models.model: 'artifact_path' is deprecated. Please use 'name' instead.
Registered model 'uas_model3' already exists. Creating a new version of this model...
Created version '4' of model 'uas_model3'.
Decision Tree trained. Run ID: 4b9db5264c4f4253ac5fa81a47aa3a2

Step 5: Evaluation

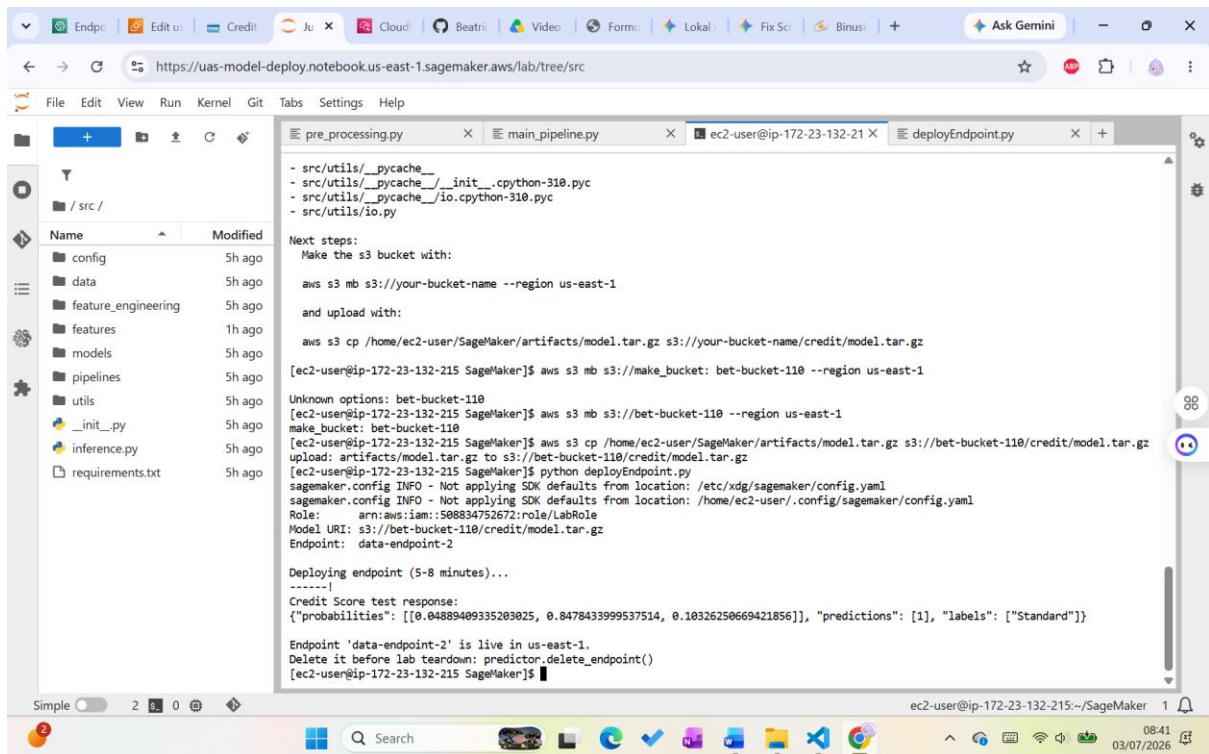
Downloading artifacts:	0%	0/1 [00:00:?, ?it/s]
Downloading artifacts:	100%	5/5 [00:00:00:00, 2292.47it/s]
Evaluation Accuracy=0.628	Precision=0.602	Recall=0.567
Downloading artifacts:	0%	0/1 [00:00:?, ?it/s]
Downloading artifacts:	100%	5/5 [00:00:00:00, 416.02it/s]
Evaluation Accuracy=0.722	Precision=0.700	Recall=0.706
Downloading artifacts:	0%	0/1 [00:00:?, ?it/s]
Downloading artifacts:	100%	5/5 [00:00:00:00, 2056.84it/s]
Evaluation Accuracy=0.681	Precision=0.659	Recall=0.639

Model Random Forest APPROVED (accuracy=0.722)
Packaged: /home/ec2-user/SageMaker/artifacts/model.tar.gz
Is! arsip saat ini:
- model_rf.joblib
- scaler.joblib
- src
- src/.ipynb_checkpoints
- src/.ipynb_checkpoints/requirements-checkpoint.txt
- src/__init__.py
- src/__pycache__
- src/__pycache__/__init__.cpython-310.pyc
- src/config
- src/config/__init__.py
- src/config/config.py

Simple 2 0

ec2-user@ip-172-23-132-215-~/SageMaker 1

08:40
03/07/2026



Note pemilihan model (penjelasan):

- Pemilihan model dilakukan dengan cara memasukkan angka secara manual kedalam model untuk melakukan tuning, hingga mendapatkan hasil model yang terbaik.

4. Dengan pengalaman anda melakukan mekanisme berbasis local dan cloud, akhirnya anda dapat membandingkan advantages dan disadvantages kedua pendekatan ini.

Advantages Cloud:

1. Website yang dibuat dapat diakses tanpa membeli hardware yang dibutuhkan.
2. Cloud sudah menyediakan url secara langsung yang dapat diakses kapanpun.
3. Memiliki kontrol penuh pada website yang digunakan tanpa harus membangun aplikasi dari dasar.
4. Jaminan uptime dan backup otomatis.

Disadvantage cloud :

1. Bergantung pada koneksi internet
2. Manajemen akses yang terlalu kompleks, aws memiliki banyak fitur kompleks terutama dalam melakukan konfigurasi keamanan jaringan virtual serta enkripsi
3. Biaya dari variabel yang digunakan dapat meledak.

Advantages local :

1. Tidak memiliki latensi karena proses membaca hingga pembuatan model yang berjalan secara langsung.
2. Biaya awal yang sudah diketahui tanpa adanya tagihan bulanan.
3. Data berada di dalam perangkat milik sendiri sehingga dapat meminimalisir resiko kebocoran data

Disadvantage local:

1. Kapasitas yang terbatas oleh spesifikasi dari laptop yang saya gunakan, sehingga crash dapat terjadi akibat kekurangan RAM atau memori.
2. Semua urusan backup dan pemulihan hardware yang rusak dilakukan sendiri.
3. Jika perangkat rusak maka file dapat hilang.

5. Penjelasan Video langkah – langkah model deployment mulai dari pembuatan notebook hingga model deployment menggunakan EC2 dan Sagemaker AI.

Github link :

<https://github.com/Beatrice-Nathania/Credit-Score-Prediction>

Video and code file Link :

https://drive.google.com/drive/folders/1DmJyjjgnH4RH_AJGF3GIBD2lpjtmNtVN?usp=drive_link

Note : beberapa folder memiliki file .init yang digunakan agar system mengenal folder sebagai library yang dapat dipanggil.

Application Link : <http://34.207.213.90:8501/>